

INTERNATIONAL SOCIETY FOR PHARMACOEPIDEMIOLOGY
PRE-CONFERENCE COURSE 2026

Let's Get Personal

Defining and Estimating Effects of Dynamic Treatment Strategies Using Real-World Data

Lecture 2

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Outline

- **Estimating the effects of dynamic treatment strategies**
IP weighting of a dynamic marginal structural model
- **Key assumptions**
- **Data requirements**
- **Alternative estimators**

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Estimation procedure

- We will focus on one of the most common approaches
 - **Inverse-probability (IP) weighting** of a **dynamic marginal structural model**
- The procedure can be summarized in **3 (simple?) steps**
 - Step 1: Construct IP weights
 - Step 2: Apply censoring rules
 - Step 3: Estimate effects via a dynamic marginal structural model

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Step 1: Construct IP weights

- Recall: Non-stabilized weights W^A for a **time-fixed treatment** A

$$W^A = \frac{1}{f(A|L)} \quad \text{If treated:} \quad W^A = \frac{1}{Pr(A = 1|L)}$$
$$\text{If untreated:} \quad W^A = \frac{1}{Pr(A = 0|L)}$$

- Denominator is (informally) the probability of having your observed treatment value A given your covariate values L

Step 1: Construct IP weights

- But we are interested in sustained, dynamic strategies
 - where treatment can vary over time
 - which means confounders can also vary over time
- We need: IP weights for a **time-varying treatment** $\bar{A}$ at time k

Final weight at time k

$$W_k^{\bar{A}} = \frac{1}{f(A_0|\bar{A}_{-1}, \bar{L}_0)} \times \frac{1}{f(A_1|\bar{A}_0, \bar{L}_1)} \times \cdots \frac{1}{f(A_k|\bar{A}_{k-1}, \bar{L}_k)} = \prod_{m=0}^k \frac{1}{f(A_m|\bar{A}_{m-1}, \bar{L}_m)}$$

Time-specific contribution

- Overbars denote history of a variable
- Subscripts represent time points

Step 1: Construct IP weights

- To estimate the weights

$$W_k^{\bar{A}} = \prod_{m=0}^k \frac{1}{f(A_m | \bar{A}_{m-1}, \bar{L}_m)}$$

- **Fit model(s) for the denominator probabilities**
 - for example, for a binary treatment, we may fit:

$$\text{logit}[\Pr(A_m = 1 | \bar{A}_{m-1}, \bar{L}_m)]$$

Step 1: Construct IP weights

- But for dynamic strategies, **special considerations** are needed
 - because treatment decisions depend on evolving patient characteristics

Step 1: Construct IP weights

- **Consideration #1**: For strategies that include **excused** times (when treatment decisions are not enforced but up to clinical discretion), weight construction needs to be modified
 - At excused times:
 - the **time-specific contribution** to the weights is **1**
 - observations are generally **excluded from the model** used to estimate denominator probabilities

Medication class can be changed if severe urinary tract infection, diabetic ketoacidosis, pancreatitis, or medullary thyroid cancer develops.

Excused

Step 1: Construct IP weights

- **Consideration #2**: The usual stabilized weights **cannot be used**
 - Because the numerator of these weights depends on an individual's *time-varying treatment*
 - See Technical Point 21.2 in Hernán MA, Robins JM (2020). *Causal Inference: What If*.
 - Alternative stabilization approaches are available (e.g., based only on baseline variables, see Cain et al. Int J Biostat. 2010)
 - but they are *not guaranteed to produce more precise estimates*

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Step 2: Apply censoring rules

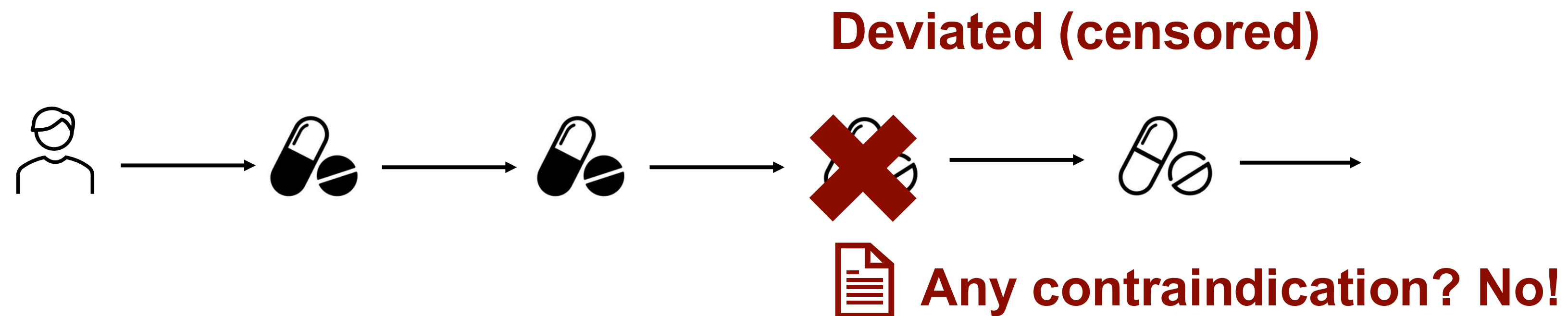
- Individuals are censored due to non-adherence when they **deviate** from their assigned treatment strategy
 - which ensures all individuals follow their assigned strategy
- Censoring rules are a function of the treatment strategies (and vice versa)
 - **deviations** from the strategy are handled via censoring
 - if **censoring** is used, the analysis **generally implies an intervention** under which the condition that triggered censoring is not allowed

Step 2: Apply censoring rules

- For dynamic strategies, **special considerations** are again needed
 - because treatment decisions (and therefore censoring rules) depend on evolving patient characteristics

Step 2: Apply censoring rules

- **Consideration #1**: Non-adherence to the strategies (deviations) can be a function of not just treatment A but also covariates L
 - individuals are censored when they deviate



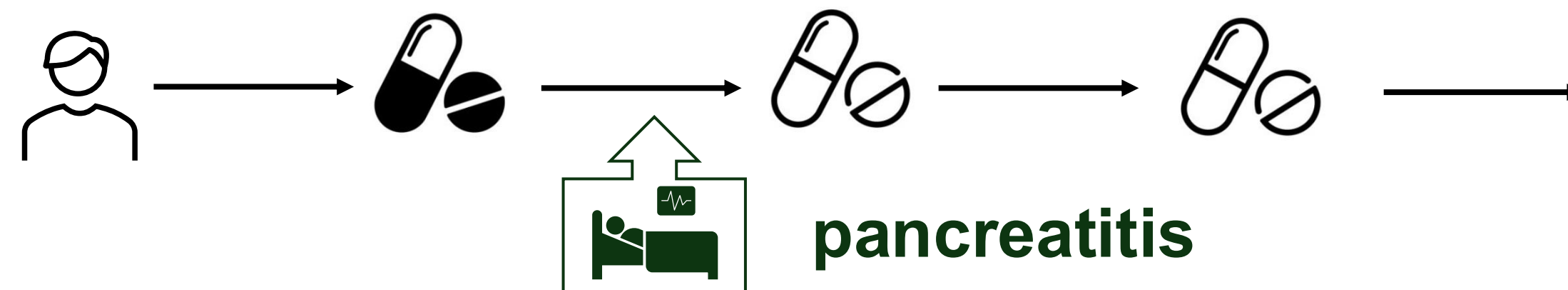
Discontinuation of treatment *prior to contraindication* is not allowed

Deviation

Step 2: Apply censoring rules

- Consideration #2: For strategies that include **excused** conditions, individuals cannot be censored if and when they are excused
 - Careful strategy specification is essential **before** censoring

Excused (NOT censored)



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Excused

Estimation procedure

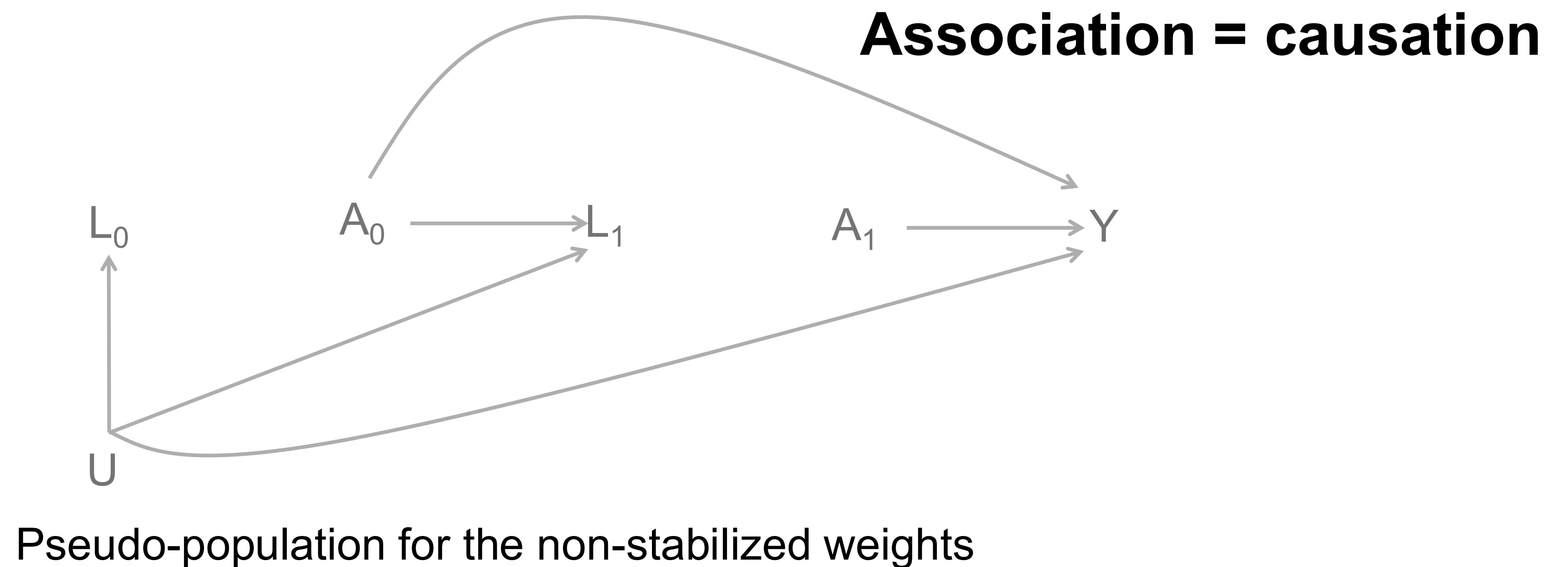
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Step 3: Estimate effects

- To estimate effects, fit an outcome model to the IP weighted and censored data
- Parameters from this model estimate parameters of a **dynamic marginal structural model (MSM)**, a *causal* model for *counterfactual outcomes*
 - Why?
 - Because in the IP weighted population, if assumptions hold
association (weighted outcome model) = causation (MSM)

Step 3: Estimate effects

- Weights create a pseudo-population in which treatment is independent of measured, time-varying confounders



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Key assumptions

- We said that, in the pseudo-population, **if assumptions hold**
association (weighted outcome model) = causation (MSM)
- That is, endowing estimates with a causal interpretation requires **assumptions**, typically the following identifying assumptions:
 - Sequential exchangeability
 - Sequential positivity
 - Sequential consistency

Key assumptions

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Key assumptions

- **Sequential exchangeability***

$$Y^g \perp\!\!\!\perp A_k | \bar{A}_{k-1} = g(\bar{A}_{k-2}, \bar{L}_{k-1}), \bar{L}_k \text{ for all strategies } g \text{ and } k = 0, 1, \dots, K$$

- the counterfactual outcome under dynamic strategy g , Y^g , is independent of treatment at every time given *past* treatment and covariates compatible with strategy g
 - i.e., no unmeasured time-fixed or time-varying confounding at any time
 - but if an individual is excused, then this assumption is no longer required!

Medication class can be changed if severe urinary tract infection, diabetic ketoacidosis, pancreatitis, or medullary thyroid cancer develops.

Excused

* For different versions of sequential exchangeability (including weaker versions that can be used to identify static strategies and stronger versions that are required when treatment strategies depend on the natural value of treatment), see *Causal Inference: What If?* Section 19.5 and Technical Point 19.3 and Sarvet and Stensrud. 2026. *Annual Review of Statistics and Its Application*.

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Key assumptions

- **Sequential positivity**

If the joint probability of a treatment and covariate history $(\bar{a}_{k-1}, \bar{l}_k)$ at time k is nonzero, then the conditional probability of receiving each type of treatment (a_k) at time k must be nonzero

- i.e., everyone has a chance to receive all levels of treatment at each time within levels of past treatment and confounders
- but if an individual is **excused**, then this assumption is no longer required!

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Key assumptions

- **Sequential consistency**

For any strategy g , if $A_k = g_k(\bar{A}_{k-1}, \bar{L}_k)$ at each time k for a given individual then $Y^g = Y$

- i.e., treatment is well-defined (only one version of treatment)
- and there is no interference (treatment status of one individual does not affect the potential outcomes of other individuals)

Stronger (generally accepted) versions of this assumption are discussed in *Causal Inference: What If?* Technical Point 19.2

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Data requirements

- Now that we know how to estimate these effects, a common question is, **“What data do we need?”**
 - In short: we need good measures of *every element* included in the target trial protocol
- This includes everything needed to define
 - eligibility criteria
 - treatment strategies (*‘assignment,’ deviations, excusing*)
 - outcomes
 - follow-up
 - covariates (generally, *all time-fixed and time-varying confounders*)

Data requirements

- **Consideration**: for dynamic treatment strategies
 - determining whether a treatment change (e.g., discontinuation) is a deviation or consistent with the strategy generally requires
 - sufficiently detailed, time-updated data on the *reasons* for treatment changes (e.g., indications, contraindications, side-effects)
- Therefore, our treatment strategies often also require **frequent measurement** (or *opportunities for measurement*) of these variables

Individuals must also have a healthcare contact at least once every 12 months

Deviation

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Alternative estimators

- Estimating effects of dynamic strategies (*almost always*) requires **g-methods**
 - that is, methods designed to handle time-varying confounding and treatment-confounder feedback

Alternative estimators

- We reviewed one g-method (**IP weighting of and MSM**)
 - but this is *not the only option*
- Alternative approaches offer trade-offs in **statistical efficiency, modeling assumptions, and robustness**

Alternative estimators

- G-methods include:
 - **Plug-in g-formula** (Robins 1986)
 - **G-estimation of structural nested models** (Robins 1989, 1991)
 - **Inverse probability weighting** (Robins 1998)
 - of a dynamic marginal structural model
 - of a dose-response marginal structural model (without censoring at non-adherence, see Toh et al. 2012 for special considerations)
 - **Doubly-robust versions** (Robins, van der Laan, Rotnitzky)
 - e.g., collaborative targeted minimum loss-based estimation, AIPW

Alternative estimators

- Methods can be extended and **combined** with other causal inference tools
 - sequential trial emulation
 - cloning
 - IP weighting for loss to follow-up
 - causal approaches for handling competing events and grace periods
 - optimal regimes
 - ... etc.
- Choice of method(s) should be guided by the question and available data

Summary

- Estimating the effects of dynamic treatment strategies requires methods that **appropriately address time-varying confounding**
 - such as IP weighting of a dynamic marginal structural model and other g-methods
- When using IP weighting of an MSM
 - weights and censoring rules need to be *tailored to* the dynamic strategy of interest
- Careful attention to **the validity of identifying assumptions** is essential
 - but dynamic strategies can also be used to *make some assumptions more plausible!*
- To estimate effects of dynamic treatment strategies, good methods aren't good enough . . . **high-quality longitudinal data** are also essential

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Thank you!

Questions?

Supplemental

- Why are MSMs “**structural**?”
 - Because they are models for counterfactual outcomes, Y^g , which are referred to as “structural” or “causal” models, e.g.,

$$E[Y^g] = \beta_0 + \beta_1 g$$

- Parameters for treatment in this structural model (β_1) have a direct causal interpretation

Supplemental

- Why are MSMs “**marginal?**”
 - Because they are models for functionals of the **marginal distribution** of the counterfactual outcome
 - They do not impose any restrictions on the joint distribution of counterfactual outcomes under different treatment strategies
 - model the mean of $Y^{g=g^1}$ and the mean of $Y^{g=g^2}$
 - but say nothing about the relationship between them*